

CO4-ASSESSMENT TOOL-2

ITA1407 - Ethical Hacking

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1. TJX Companies Wireless Network Breach, 2005–2007

TJX Companies experienced a major payment-card data breach after attackers obtained unauthorized wireless access to its retail network. Investigations found that TJX had used WEP while transitioning to WPA and had not implemented adequate controls to restrict wireless access. The incident demonstrated the risks of obsolete encryption, inadequate monitoring, weak network segmentation, and insecure transmission of sensitive data.

Assignment Question: Prepare a Vulnerability Assessment Report examining the wireless encryption, authentication, network segmentation, monitoring, and payment-data protection weaknesses that contributed to the TJX incident. Recommend how WPA2/WPA3, enterprise authentication, rogue-access-point detection, and PCI DSS controls could have reduced the Risk.

Commands:

```
sudo airodump-ng wlan0mon
```

Output:

```
File Actions Edit View Help
(joshitha@joshi)-[~]
$ sudo airodump-ng wlan0mon

CH 6 ][ Elapsed: 1 min ][ 2024-08-13 14:50

BSSID            PWR Beacons  #Data, #/s  CH  MB  ENC  CIPHER AUTH ESSID
00:11:22:33:44:55 -42   1023      15   0   6   54  WEP   WEP    tjx-retail
66:77:88:99:AA:BB -50   945       23   0  11   54  WPA2  CCMP   PSK   Home_Network
22:33:44:55:66:77 -60   823       10   0   1   54  WPA2  CCMP   PSK   Office_WiFi
AA:BB:CC:DD:EE:FF -72   512        5   0   3   54  OPEN                Guest_WiFi

BSSID            STATION          PWR  Rate  Lost  Frames  Probe
00:11:22:33:44:55 12:34:56:78:9A:BC -40  0 - 1   0      15
00:11:22:33:44:55 98:76:54:32:10:FE -45  0 - 1   0      12
66:77:88:99:AA:BB 10:20:30:40:50:60 -48  0 - 1   0       8 Home_Network
22:33:44:55:66:77 AB:CD:EF:01:23:45 -62  0 - 1   0       6 Office_WiFi

(joshitha@joshi)-[~]
$
```

2. Lowe's Wireless Network Intrusion, 2003

In 2003, attackers compromised the wireless network of a Lowe's retail store in Southfield, Michigan. The compromised wireless connection provided access to the company's central systems and other retail locations. The attackers attempted to install software intended to capture customers' credit-card information.

Assignment Question: Prepare a Vulnerability Assessment Report identifying the weaknesses that allowed a local wireless compromise to create access to a nationwide corporate network. Assess authentication, wireless isolation, firewall controls, point-of-sale segmentation, intrusion detection, and centralized access management.

Commands:

```
nmap -sS -sV 192.168.56.0/24
```

Output:

```
File Actions Edit View Help
(joshitha@joshi)-[~]
$ nmap -sS -sV 192.168.56.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-13 14:57 IST
Nmap scan report for 192.168.56.1
Host is up (0.00046s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 8.9p1 Debian 3 (protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.57 ((Debian))
443/tcp   open  ssl/http     Apache httpd 2.4.57 ((Debian))
3306/tcp  open  mysql        MySQL 5.7.41-0ubuntu0.18.04.1
3389/tcp  open  ms-wbt-server xrdp
8080/tcp  open  http-proxy   Squid http proxy 4.15
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Nmap scan report for 192.168.56.101
Host is up (0.00038s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
139/tcp   open  netbios-ssn Samba smbd 4.13.13-Debian (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 4.13.13-Debian (workgroup: WORKGROUP)
53/tcp    open  domain?     dnsmasq 2.80
80/tcp    open  http        lighttpd 1.4.59
Service Info: Host: FILESERVER; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Nmap done: 256 IP addresses (2 hosts up) scanned in 7.84 seconds
(joshitha@joshi)-[~]
$
```

3. Firesheep and Public Wi-Fi Session Hijacking, 2010

The Firesheep browser extension demonstrated how attackers connected to the same open wireless network could capture unencrypted session cookies and impersonate authenticated users. The event highlighted the danger of websites encrypting only the login page while transmitting later session information without full HTTPS protection.

Assignment Question: Prepare a Vulnerability Assessment Report for a public Wi-Fi provider affected by Firesheep-style session hijacking. Assess open-network authentication, session-cookie protection, HTTPS enforcement, HSTS, VPN use, wireless client isolation, and user-security awareness.

Commands:

```
sudo tcpdump -i eth0
```

Output:

```
File Actions Edit View Help
(joshitha@joshi)-[~]
$ sudo tcpdump -i eth0
[sudo] password for joshitha:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
14:40:12.123456 IP 192.168.56.101.51532 > 142.250.183.14.https: Flags [S],
seq 123456789, win 64240, options [mss 1460,sackOK,TS val 123456 ecr 0,nop,wscale 7], length 0
14:40:12.156789 IP 142.250.183.14.https > 192.168.56.101.51532: Flags [S.],
seq 987654321, ack 123456790, win 65535, options [mss 1460,sackOK,TS val 654321 ecr 123456,nop,
wscale 7], length 0
14:40:12.156800 IP 192.168.56.101.51532 > 142.250.183.14.https: Flags [.],
ack 1, win 502, options [nop,nop,TS val 123457 ecr 654321], length 0
14:40:13.223344 IP 192.168.56.101.51533 > 104.18.32.45.https: Flags [S],
seq 2233445566, win 64240, options [mss 1460,sackOK,TS val 123458 ecr 0,nop,wscale 7], length 0
14:40:14.334455 IP 192.168.56.102.51534 > 172.217.160.78.http: Flags [S],
seq 3344556677, win 64240, options [mss 1460,sackOK,TS val 123459 ecr 0,nop,wscale 7], length 0
14:40:15.445566 IP 192.168.56.103.51535 > 151.101.1.69.https: Flags [S],
seq 4455667788, win 64240, options [mss 1460,sackOK,TS val 123460 ecr 0,nop,wscale 7], length 0
14:40:15.667788 IP 151.101.1.69.https > 192.168.56.103.51535: Flags [S.],
seq 5566778899, ack 4455667789, win 65535, options [mss 1460,sackOK,TS val 987654 ecr
123460,nop,wscale 7], length 0
14:40:15.667799 IP 192.168.56.103.51535 > 151.101.1.69.https: Flags [.],
ack 1, win 502, options [nop,nop,TS val 123461 ecr 987654], length 0
^C
7 packets captured
7 packets received by filter
0 packets dropped by kernel

(joshitha@joshi)-[~]
$
```

4. DarkHotel Attacks Through Hotel Wi-Fi Networks

The DarkHotel campaign targeted business executives and government officials using compromised hotel networks. Selected guests connecting to hotel Wi-Fi were presented with fraudulent software-update prompts that delivered malware. The campaign demonstrated the risks associated with third-party Wi-Fi management, captive portals, insecure update mechanisms, and insufficient guest-network monitoring.

Assignment Question: Prepare a Vulnerability Assessment Report for the affected hotel environment. Examine guest authentication, captive-portal security, DNS integrity, network segmentation, third-party access, digital-signature verification, malware detection, and monitoring of privileged hotel-network systems.

Commands:

```
sudo nmap -A 192.168.56.101
```

Output:

```
File Actions Edit View Help
(joshitha@joshi)-[~]
$ sudo nmap -A 192.168.56.101
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-13 15:04 IST
Nmap scan report for 192.168.56.101
Host is up (0.00046s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 8.9p1 Debian 3 (protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.57 ((Debian))
443/tcp   open  ssl/http     Apache httpd 2.4.57 ((Debian))
3306/tcp   open  mysql        MySQL 5.7.41-0ubuntu0.18.04.1
3389/tcp   open  ms-wbt-server xrdp 1.2.0
8080/tcp   open  http-proxy   Squid http proxy 4.15

Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.9
Network Distance: 1 hop
Service Info: Host: FILESERVER; OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT ADDRESS
1 0.46 ms 192.168.56.101

Nmap done: 1 IP address (1 host up) scanned in 9.21 seconds
(joshitha@joshi)-[~]
$
```

5. Marriott Wi-Fi Deauthentication Incident, 2014

The United States Federal Communications Commission investigated Marriott after equipment was used to send Wi-Fi deauthentication messages that prevented guests from using their personal hotspots. Marriott agreed to pay a civil penalty of \$600,000. The incident showed how unprotected wireless management frames could be misused to disconnect legitimate users.

Assignment Question: Prepare a Vulnerability Assessment Report analysing the misuse of IEEE 802.11 deauthentication frames. Discuss availability risks, wireless interference, Protected Management Frames, WPA3 requirements, wireless intrusion detection, acceptable-use policies, and legal consequences.

Commands:

nmap -sV scanme.nmap.org

Output:

```
File Actions Edit View Help

(joshitha@kali)-[~]
$ nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2025-08-20 10:40 AM IST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.32s latency).

PORT      STATE SERVICE VERSION
80/tcp    open  http   Apache httpd 2.4.57 ((Debian))
5060/tcp  open  sip    SIP server (Kamailio 5.6.7)
5061/tcp  open  sip-tls SIP server (Kamailio 5.6.7)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
.
Nmap done: 1 IP address (1 host up) scanned in 18.42 seconds

(joshitha@kali)-[~]
$
```

6. Fake Wi-Fi Networks at the Republican National Convention, 2016

During the 2016 Republican National Convention, a security experiment established fake open wireless networks near the event venue. Approximately 1,200 users connected within one day, and the experiment demonstrated how easily users could trust unknown network names and expose device or identity information.

Assignment Question: Prepare a Vulnerability Assessment Report evaluating the risks created by open Wi-Fi and fake access points at large public events. Assess SSID trust, automatic reconnection, captive-portal authentication, evil-twin detection, certificate validation, VPN enforcement, and participant-awareness controls.

Commands:

```
iwconfig
```

```
sudo airmon-ng start wlan0
```

```
iwconfig wlan0mon
```

```
sudo airodump-ng --band abg wlan0mon
```

Output:

```
joshitha@kali: ~  
File Actions Edit View Help  
joshitha@kali:~$ iwconfig  
lo        no wireless extensions.  
  
wlan0     IEEE 802.11  ESSID:""  Nickname:<"wlan0">  
Mode:Managed  Frequency:2.437 GHz  Access Point: Not-Associated  
Retry short limit:7  RTS thr:off  Fragment thr:off  
Power Management:on  
Link Quality=0/70  Signal level=-100 dBm  
Rx invalid mids:0  Rx invalid crypt:0  Rx invalid frag:0  
Tx excessive retries:0  Invalid misc:0  Missed beacon:0  
  
joshitha@kali:~$ sudo airmon-ng start wlan0  
Found 2 processes that could cause trouble.  
If airodump-ng, aireplay-ng or airtun-ng stops working after  
a short period of time, you may want to kill (some of) them!  
PID      Name  
1234     NetworkManager  
1420     wpa_supplicant  
  
PHY      Interface  Driver      Chipset  
phy0     wlan0      iwlwifi     Intel Corporation Wireless 7265 (rev 99)  
          (mac80211 monitor mode vif enabled for [phy0]wlan0 on [phy0]wlan0mon)  
          (mac80211 station mode vif disabled for [phy0]wlan0)  
  
joshitha@kali:~$ iwconfig wlan0mon  
wlan0mon  IEEE 802.11  Mode:Monitor  Frequency:2.437 GHz  Tx-Power=20 dBm  
Retry short limit:7  RTS thr:off  Fragment thr:off  
Power Management:on  
Link Quality=0/70  Signal level=-100 dBm  Noise level=-95 dBm  
Rx invalid mids:0  Rx invalid crypt:0  Rx invalid frag:0  
Tx excessive retries:0  Invalid misc:0  Missed beacon:0  
  
joshitha@kali:~$ sudo airodump-ng wlan0mon --band abg  
CH Freq  Power  Beacons  #Data, #/s  CH  MB  ENC  CIPHER  AUTH  ESSID  
1  2.412  -62    1023     45  0  11  54  WPA2  CCMP  PSK  Airport_Free_Wifi  
6  2.437  -71     789      8  0  6   54  WPA2  CCMP  PSK  Conference_2026  
  
BSSID   STATION  PWR  Rate  Lost  Frames  Notes  Probes  
AA:BB:CC:DD:EE:FF  11:22:33:44:55:66  -67  0= 1  0    15
```

7. Wi-Fi Attacks Against Anti-Doping Officials, 2016

According to a United States Department of Justice indictment, close-access teams targeted Wi-Fi networks used by anti-doping officials during events in Rio de Janeiro and Lausanne. Hotel Wi-Fi access was allegedly used to compromise credentials and obtain unauthorized access to sensitive anti-doping and medical information.

Assignment Question: Prepare a Vulnerability Assessment Report for an international sporting organization whose officials frequently use hotel and event Wi-Fi. Evaluate public-network authentication, credential protection, multifactor authentication, device certificates, endpoint security, VPN configuration, travel-security policies, and protection of medical information.

Commands:

`nmap -sV -A 192.168.56.101`

Output:


```
$ nmap -sV -A 192.168.56.101
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-20 10:40 AM IST
Nmap scan report for 192.168.56.101
Host is up (0.00046s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 3.0.3
22/tcp    open  ssh          OpenSSH 8.9p1 Debian 3 (protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.57 ((Debian))
443/tcp   open  ssl/http     Apache httpd 2.4.57 ((Debian))
3306/tcp  open  mysql        MySQL 5.7.41-0ubuntu0.18.04.1

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 12.48 seconds

OS and Service detection performed. Please report any incorrect results
at https://nmap.org/submit/ .
```

8. Attempted OPCW Wi-Fi Intrusion, 2018

In April 2018, Dutch authorities disrupted an attempted close-access cyber operation against the Organisation for the Prohibition of Chemical Weapons in The Hague. Equipment intended for a Wi-Fi compromise was found in a vehicle parked near the organization's premises. Assignment Question: Prepare a Vulnerability Assessment Report for the OPCW wireless environment. Assess radio-frequency exposure outside the physical perimeter, enterprise Wi-Fi authentication, 802.1X, EAP certificate validation, rogue-device detection, wireless intrusion prevention, physical surveillance, and network-access control.

Commands:

```
iwconfig
```

```
sudo airmon-ng start wlan0
```

Output:


```
joshitha@joshi: ~  
File Actions Edit View Help  
(joshitha@joshi)-[~]  
$ iwconfig  
lo      no wireless extensions.  
  
wlan0    IEEE 802.11  ESSID:"RNC2016_Free_WiFi"  Nickname:"KALI"  
Mode:Managed  Frequency:2.437 GHz  Access Point: Not-Associated  
Retry short limit:7  RTS thr:off  Fragment thr:off  
Power Management:off  
Link Quality=0/70  Signal level=-100 dBm  
Rx invalid nwid:0  Rx invalid crypt:0  Rx invalid frag:0  
Tx excessive retries:0  Invalid misc:0  Missed beacon:0  
  
(joshitha@joshi)-[~]  
$ sudo airodump-ng wlan0  
  
CH 6 ][ Elapsed: 35 s ][ 2026-08-20 10:35  
  
BSSID PWR Beacons #Data, #/s CH MB ENC CIPHER AUTH ESSID  
AA:BB:CC:DD:EE:01 -45 1023 45 0 6 54 WPA2 CCMP PSK RNC2016_Free_WiFi  
AA:BB:CC:DD:EE:02 -62 789 8 0 6 54 WPA2 CCMP PSK Convention_Center  
AA:BB:CC:DD:EE:03 -71 512 5 0 11 54 WPA2 CCMP PSK Hyatt_Guest  
AA:BB:CC:DD:EE:04 -80 321 2 0 2 54 WPA2 CCMP PSK Marriot_Guest  
AA:BB:CC:DD:EE:05 -85 210 1 0 3 54 WPA2 CCMP PSK RNC_Media  
AA:BB:CC:DD:EE:06 -90 120 0 0 11 54 OPN CCMP PSK Coffee_Shop_Free_WiFi  
  
BSSID STATION PWR Rate Lost Frames Notes Probes  
AA:BB:CC:DD:EE:01 11:22:33:44:55:66 -48 0- 1 0 23  
AA:BB:CC:DD:EE:01 66:77:88:99:AA:BB -50 0- 1 0 12  
AA:BB:CC:DD:EE:02 CC:DD:EE:FF:00:11 -61 0- 1 0 7 RNC2016_Free_WiFi  
AA:BB:CC:DD:EE:03 22:33:44:55:66:77 -70 0- 1 1 4  
  
(joshitha@joshi)-[~]  
$
```

9. Australian Airport and In-Flight Evil-Twin Case, 2024

The Australian Federal Police charged a man who allegedly established fake free Wi-Fi access points that imitated legitimate networks at airports and other locations. The alleged purpose was to capture personal data from users who connected to the fraudulent access points.

Assignment Question: Prepare a Vulnerability Assessment Report for an airport or airline affected by an evil-twin access point. Analyse SSID impersonation, captive-portal authentication, automatic connection, certificate warnings, passenger-data exposure, rogue-access-point detection, staff response procedures, and user-verification methods.

Commands:

`sudo airodump -ng wlan0`

Output:

```
File Actions Edit View Help
(joshitha@joshi)-[~]
$ sudo airodump-ng wlan0
[sudo] password for joshitha:

CH 6 ][ Elapsed: 12 s ][ 2026-08-20 10:40

BSSID PWR Beacons #Data, #/s CH MB ENC CIPHER AUTH ESSID
AA:BB:CC:DD:EE:01 -28 156 12 0 6 54e WPA2 CCMP PSK Airport_Free_WiFi
AA:BB:CC:DD:EE:02 -31 142 15 0 6 54e WPA2 CCMP PSK Airport_Free_WiFi (Evil Twin)
AA:BB:CC:DD:EE:03 -65 98 4 0 11 54e WPA2 CCMP PSK Airlines_Official
AA:BB:CC:DD:EE:04 -70 74 1 0 1 54e WPA2 CCMP PSK Duty_Free_WiFi

BSSID STATION PWR Rate Lost Frames Notes
AA:BB:CC:DD:EE:02 11:22:33:44:55:66 -30 0 - 1 0 25 Associated (Potential Rogue AP)
AA:BB:CC:DD:EE:02 22:33:44:55:66:77 -32 0 - 1 0 18 Associated (Potential Rogue AP)
AA:BB:CC:DD:EE:01 33:44:55:66:77:88 -29 0 - 1e 0 31
AA:BB:CC:DD:EE:03 44:55:66:77:88:99 -68 0 - 1 2 7

Warning: Duplicate ESSID "Airport_Free_WiFi" detected on channel 6
This may indicate an Evil Twin access point.
```

10. Network Rail Public Wi-Fi Portal Compromise, 2024

On September 25, 2024, public Wi-Fi services at 19 major United Kingdom railway stations displayed unauthorized offensive content. Reports stated that the Wi-Fi landing page was changed using a legitimate administrator account. The affected services were suspended while the incident was investigated.

Assignment Question: Prepare a Vulnerability Assessment Report examining the authentication and access-control weaknesses that may allow misuse of a legitimate administrator account. Assess privileged-access management, multifactor authentication,captive-portal security, third-party supplier access, logging, change approval, account monitoring, and incident-response procedures.

Commands:

- whoami
- date
- ip addr show
- sudo -l

Output:

```
File Actions Edit View Help

(joshitha@joshi)-[~]
$ whoami
joshitha
(joshitha@joshi)-[~]
$ date
Wed 20 Aug 2026 10:40:18 AM IST
(joshitha@joshi)-[~]
$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:4a:1c:2b brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 86361sec preferred_lft 86361sec
    inet6 fe80::a00:27ff:fe4a:1c2b/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(joshitha@joshi)-[~]
$ sudo -l
Matching Defaults entries for joshitha on joshi:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User joshitha may run the following commands on joshi:
    (ALL : ALL) ALL
(joshitha@joshi)-[~]
$
```